

2 **(long)** Figure 2 shows the plan view of a channel formed during a dam break. The bottom of the channel is at the same height as the bottom of the reservoir. The water surface in the reservoir has height h_0 , which can be assumed to remain constant. The water in the channel has width $b(x)$, surface height $h(x)$, and speed $v(x)$. The streamline curvature can be neglected, meaning that all flow properties vary only in the x -direction. For this problem, a non-dimensional number is defined as $Fr = v/\sqrt{gh}$, where g is the acceleration due to gravity.

(a) Explain why, at a given x -location, the pressure variation in the vertical z -direction is hydrostatic: i.e. $p(z) + \rho gz = 0$. [2]

(b) If friction with the channel walls can be ignored and there is no sudden increase in h (known as a hydraulic jump), explain why Bernoulli's equation can be used to relate $h(x)$ and $v(x)$ throughout the flow with $v^2 + 2gh = \text{const.}$ [4]

(c) Properties at the narrowest section of the channel are labelled with subscript 1. Given knowledge that $Fr_1 = 1$, find the simplest possible expressions for h_1 and v_1 in terms of g and h_0 , and find an expression for the volumetric flowrate, Q , through the channel in terms of g , h_0 , and b_1 . [6]

(d) Downstream of the narrowest section, the depth $h(x)$ continues to drop. What happens to the speed, $v(x)$? What is the maximum speed that the flow can attain? [2]

(e) Further downstream there is a hydraulic jump through which the height $h(x)$ suddenly increases from h_2 to h_3 and Fr changes from Fr_2 to Fr_3 . The jump is sufficiently short that you may assume b to be uniform across the jump. Defining $H = h_3/h_2$ and considering a control volume around the hydraulic jump,

(i) use conservation of mass to show that [2]

$$Fr_2^2 = H^3 Fr_3^2$$

(ii) use the steady flow momentum equation to show that [8]

$$2Fr_2^2 + 1 = H^2(2Fr_3^2 + 1)$$

(f) Check that the above equations are satisfied if there is no jump ($H = 1$). Factorize out this solution and then find the other two solutions for H in terms of Fr_2 . State which solution is unphysical. What is H if $Fr_2 = \sqrt{3}$? [6]

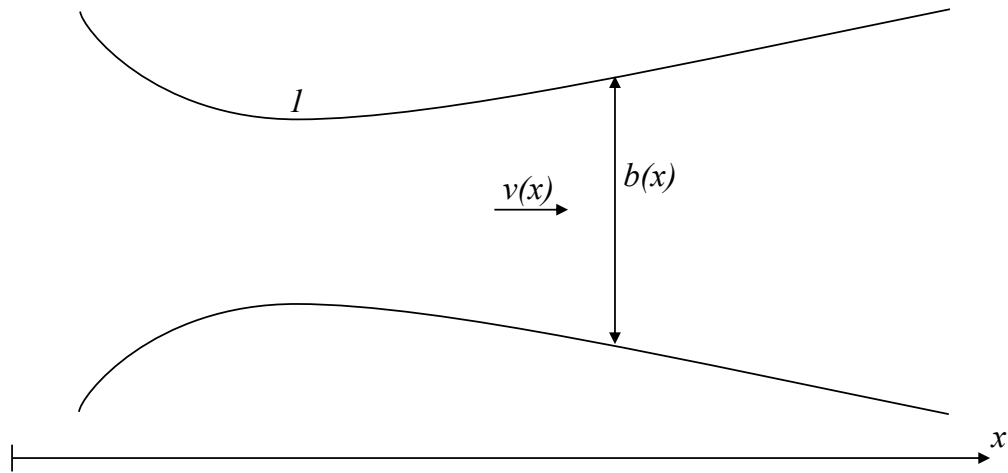


Fig. 2